Variable

A variable is a container that stores data in a program. we can use the variable name to access.

```
In [ ]: #Creating variables
        a=10 # variable = value
```

```
In [ ]: #string
        name = "Reeta"
        print(name)
        print(type(name))
```

```
In [ ]: #integer
        age = 20
        print(age)
        print(type(age))
```

```
In [ ]: #float
        fee=20000.00
        print(fee)
        print(type(fee))
```

```
In [ ]: #boolean
        result = False
        print(result)
        print(type(result))
```

```
In [ ]: #changing variabl value
        age = 20
        age = 21
        print(age)
```

```
In [ ]: #multiple variable
        name,age = "Reeta",20
        print(name)
        print(age)
```

```
In [ ]: #assign same value
        a=b=c=10
        print(a)
        print(b)
        print(c)
```

Casting = Casting means converting one data type to another data type.

```
In [ ]: # str to int
        age="20"
        age=int(age)
        print(type(age))
        #boolean all num comes True(postive , negative)expect 0 and Flase(zero).
```

```
In [14]: # float to int
         CGPA = 8.51
         per =int(CGPA)
         print(per)
```

```
8
```

```
In [12]: #boolean to int
         Grade = True
         rank = int(Grade)
         print(rank)
```

```
1
```

Naming Conventions

Rules for naming variables, functions, and other objects in Python.

Types:

1. Camel Case

2. Pascal Case

3. Snake Case

```
In [19]: #1. Camel Case: First word starts with a small letter; other words start with capital letters.
         myName = "Reeta"
         print(myName)
```

```
Reeta
```

```
In [23]: #2. Pascal Case: Every word starts with a capital letter.
         MyName = "Reeta"
         print(MyName)
```

```
Reeta
```

```
In [25]: #3. Snake Case: Words are separated by underscores (_).
         my_name = "Reeta"
         print(my_name)
```

```
Reeta
```

Variable Scope

Local variable: Created inside a function.

Global variable: Created outside a function.

```
In [44]: num1 = 10 #global
         def sum():
             num2 = 20 #local
             print(num1 + num2)
         sum()
```

```
30
```

```
In [33]: #Variable Reassignment
         a = 20
         a = 22
         print(a)
```

```
22
```

```
In [27]: #Deleting Variables (del)
         num1 = 10
         num2 = 20
         del num1
         print(num1)
         print(num2)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[27], line 5
      3 num2 = 20
      4 del num1
----> 5 print(num1)
      6 print(num2)

NameError: name 'num1' is not defined
```

```
In [29]: num1 = 10
         num2 = 20
         del num1
         print(num2)
```

```
20
```

```
In [52]: #Object Identity (id()) - The id() function returns an object's identity.
         a = 10
         b = 20
         print(id(a))
         print(id(b))
```

```
140724664543960
140724664544280
```

```
In [54]: #Unpacking Variables
         #Assigning multiple values to multiple variables in one line
         a, b, c = 10, 20, 30

         print(a)
         print(b)
         print(c)
```

```
10
20
30
```

Practice Questions

1.Write a program to swap two variables using a third variable.

```
In [13]: num1 = int(input("enter num1:"))
         num2 = int(input("enter num2:"))
         num3 = num1
         num1 = num2
         num2 = num3
         print("num1:",num1)
         print("num2:",num2)
```

```
num1: 20
num2: 10
```

2.Write a program to swap two variables without using a third variable.

```
In [5]: num1 = int(input("enter num1:"))
        num2 = int(input("enter num2:"))
        num1 , num2 = num2 , num1
        print("num1:",num1)
        print("num2:",num2)
```

```
num1: 67
num2: 23
```

3.Write a program to find the sum of two numbers stored in variables.

```
In [16]: num1 = int(input("enter num1:"))
         num2 = int(input("enter num2:"))
         sum = num1 + num2
         print("sum=",sum)
```

```
sum: 50
```

4.Write a program to print the data type of different variables.

```
In [4]: a=int(input("integer:"))
        print(type(a))
        b=float(input("float:"))
        print(type(b))
        c=input("String:")
        print(type(c))
        d=bool(input("boolean:"))
        print(type(d))
        e=complex(input("complex:"))
        print(type(e))
```

```
<class 'int'>
<class 'float'>
<class 'str'>
<class 'bool'>
<class 'complex'>
```

5.Write a program to assign the same value to multiple variables in one line.

```
In [7]: a=b=c=d=10
        print(a)
        print(b)
        print(c)
        print(d)
```

```
10
10
10
10
```

6.Write a program to assign different values to multiple variables in one line.

```
In [10]: name , age , rank = "Reeta" , 20,True
         print("name:",name)
         print("age=",age)
         print("rank:",rank)
```

```
name: Reeta
age= 20
rank: True
```

7.Write a program to convert a string variable to an integer and add 10 to it.

```
In [17]: num1 = input("enter str:")
         num2 = int(num1)
         sum = num2 + 10
         print("sum:",sum)
```

```
sum: 30
```

8.Write a program to show the difference between a local and a global variable.

```
In [26]: #local - access inside code
         def sum():
             x=10
             y=20
             print(x+y)
         sum()
```

```
30
```

```
In [28]: #global - access anywhere in the code
         x=10
         def sum():
             y=20
             print(x+y)
         sum()
```

```
30
```

9.Write a program to change a global variable inside a function.

```
In [33]: num1 = 20
         def fun():
             global num1
             num1 = 10
             print(num1)
         fun()
```

```
10
```

10.Write a program to check if two variables have the same memory address.

```
In [40]: num1 = 10
         num2 = 10
         print(num1 is num2)
```

```
True
```

In [42]:
```
num1 = 10
num2 = num1
print(num1 is num2)
```

```
True
```